RISHANTH RAJENDHRAN

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EDUCATION

University of Maryland
College Park, Maryland, US
PhD in Computer Science
Advisor: Mohit Iyyer
Jan 25 - present

University of Massachusetts
Amherst, Massachusetts, US
PhD in Computer Science
Advisor: Mohit Iyyer
Aug 24 - Dec 24

University of Utah
Salt Lake City, Utah, US
Master of Science in Computer Science
Advisor: Ana Marasovic
Aug 22 - May 24

IIT Madras
Chennai, Tamil Nadu, India
Bachelor of Technology in Computer Science and Engineering
Jul 18 - May 22

PUBLICATIONS

1. **Rishanth Rajendhran**, Amir Zadeh, Matthew Sarte, Chuan Li, Mohit Iyyer. *VeriFastScore: Speeding up long-form factuality evaluation*. Preprint
2. Ashim Gupta, **Rishanth Rajendhran**, Nathan Stringham, Vivek Srikumar, Ana Marasović. *Whispers of Doubt Amidst Echoes of Triumph in NLP Robustness*. NAACL '24

SKILLS

Programming/Scripting Languages	Python, C, C++, Javascript, SQL, OWL, HTML, CSS
Frameworks/Tools/Libraries	HuggingFace, PyTorch, Pandas, Numpy, Tensorflow, ReactJS, Git, wandb
Natural Languages	Tamil, English

RESEARCH EXPERIENCES

Research Assistant
CLIP at UMD
College Park, Maryland, US
Jan 25 - Present

Supervisor: Prof. Mohit Iyyer

VeriFastScore: Evaluating factuality of long-form LLM generations
CoTLM: Teaching LLMs to reason about next token

Research Assistant
UMassNLP
Amherst, Massachusetts, US
Aug 24 - Dec 24

Supervisor: Prof. Mohit Iyyer

Research Assistant
UtahNLP
Salt Lake City, Utah, US
May 23 - Aug 23

Supervisor: Prof. Ana Marasovic

RLHF for NLP: Improved free-text explanations generated by large-scale language models using fine-grained reinforcement learning with human feedback (FG RLHF)
Expressing model uncertainties: Explored introducing model uncertainties in self-rationalization techniques introduced in STaR Self-Taught Reasoner and ZARA Zero-shot Augmentation of Rational-Answer pairs

Student Researcher — *Salt Lake City, Utah, US*
UtahNLP — Nov 22 - May 23

Supervisor: Prof. Ana Marasovic

Evaluated robustness of large-scale language models on contrastive and out-of-domain datasets
Experimented with evaluation toolkits such as llm-foundry and lm-eval-harness

Undergraduate Researcher — *Chennai, Tamil Nadu, India*
SENAI, IIT Madras — Aug 21 - Feb 22

Supervisor: Prof. Raghunathan Rengaswamy

Worked on evolving neural networks and constrained models
Investigated autoML and EigenGame

TEACHING EXPERIENCES

Teaching Assistant — *Salt Lake City, Utah, US*
University of Utah — Jan 24 - May 24

Supervisor: Prof. Ana Marasovic
Course: CS 6340 / 5340 Natural Language Processing

Teaching Assistant — *Salt Lake City, Utah, US*
University of Utah — Aug 23 - Dec 23

Supervisor: Prof. Anna Fariha
Course: CS 6353 / 5353 Deep Learning

Teaching Assistant — *Salt Lake City, Utah, US*
University of Utah — Jan 22 - May 23

Supervisor: Prof. Daniel Kopta
Course: CS 5350 Database Management Systems

Teaching Assistant — *Salt Lake City, Utah, US*
University of Utah — Aug 22 - Dec 23

Supervisor: Prof. Ellen Riloff
Course: CS 6340 / 5340 Natural Language Processing

Instructor — *Chennai, Tamil Nadu, India*
Shaastra Tech Fest, IIT Madras — Jan 22

Conducted two-day workshop on python programming and web and app development

Exam Coach — *Chennai, Tamil Nadu, India*
Kendriya Vidhyalaya, IIT Madras — Feb 20

Volunteered to teach programming in C++ to 12th grade students from underprivileged backgrounds
Held theory followed by lab sessions to solidify understanding

OTHER EXPERIENCES

Web Developer Intern *Remote*
EduVisory, Nirmaan-IITM incubated startup Jun 20 - Jul 20; Aug 20 - Jan 21

Supervisor: Dr. Ravi Dadsena

- Worked on building website for this ed-tech startup from scratch
- Designed user interface of website and assisted the development of backend

App Developer *Remote*
Product Design Club, Centre for Innovation, IIT Madras Aug 20 - Apr 21

- Worked on building website and app for streamlining garbage disposal inside the IIT Madras campus

Coordinator *Remote*
Institute WebOps, IIT Madras Jun 20 - Apr 21

- Worked on maintaining institute websites
- Organized learning tasks for onboarding students

FEATURED PROJECTS

C-STaR Fall 24
University of Massachusetts Amherst
Github: https://github.com/vule20/C-STaR

- Incorporating uncertainty in STaR pipeline to improve model-generated reasoning

Salience for Automatic Speech Recognition Fall 23
University of Utah
Github: https://github.com/jacobkj314/salASR

- Extended input attribution techniques found in deep learning literature to the task of ASR
- Verified the soundness of salience as an explainability method for Whisper models using remove-and-retrain (RoaR) approach

Relation Extraction using Distant Supervision Spring 23
University of Utah
Github: https://github.com/RishanthRajendhran/relationExtraction

- Constructed a dataset from Wikipedia using relations in FB15K, a knowledge base completion dataset
- Built a BERT-based model for 10-class classification
- Built a Tree LSTM-based model for 10-class classification

Robustness Evaluation Fall 22 - Fall 23
University of Utah
Github: https://github.com/RishanthRajendhran/robustnessEvaluation

- Evaluated SotA LLMs on contrast sets
- Explored greedy-decoding and self-consistency with and without chain-of-thoughts

Visual Question Answering Spring 23
University of Utah
Github: https://github.com/RishanthRajendhran/visualQA

- Built an encoder-decoder model with RESNET as the base

EigenGame Fall 21 - Spring 22
IIT Madras
Github: https://github.com/RishanthRajendhran/EigenGame

Replicated results from the paper that introduced EigenGame
Conducted experiments and investigated effects of various initializations and gradient ascent approaches using 3D visualizations and plots

Transliteration using RNNs Spring 21
IIT Madras
Github: https://github.com/RishanthRajendhran/transliterationUsingRNNs

Built a Tamil-to-English transliteration system
Investigated attention mechanism using gradient-based attribution by visualizing attention heatmaps
Identified weaknesses of the model by inspection of plots

CERTIFICATIONS

Amazon ML Summer School *Online*
July 2021

Interacted with Amazon ML Engineers; discussed practical issues with using ML models on real world applications

NVIDIA Machine Learning and Deep Learning *Online*
June 2021

Workshop on choosing the right ML model for given data, transformer architecture. based Natural Language Processing applications
Course covered machine learning models and techniques